Shivang Chopra

Ph.D. in Computer Science at Georgia Institute of Technology

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Google Scholar:

Research Interests

Robust Representation Learning: Investigating the algorithmic and geometric principles underlying generalization and efficiency in multimodal foundation models, with a focus on transformer architecture design, and representation structure.

Adaptive Computation and Efficient Inference: Developing methods to scale virtual depth and adaptive computation at inference time, enabling pretrained LLMs to achieve deeper reasoning capabilities without increasing total parameter counts.

Data-Efficient Learning and Training Optimization: Designing data selection, active learning, and manifold-aware training methods to improve sample efficiency, training stability, and generalization in large-scale multimodal and language models.

Education

Georgia Institute of Technology — Jan 2025 - May 2029 (expected)
Ph.D. in Computer Science
Advisor: Prof. Zsolt Kira
Publications: CVPR 2026 [2], CVPR(W) 2025 [8], CVPR 2025 [4]

Georgia Institute of Technology — Aug 2022 - Dec 2024
Master of Science in Computer Science — **CGPA:** 3.9 / 4.0
Thesis: CRAFT: Curriculum Rank Adversarial Fine-Tuning for Robust Vision Language Models
Publications: WACV 2025 [3], CoRL 2023 [6], ICML(W) 2022 [10]

Delhi Technological University — Aug 2016 - May 2020
Bachelor of Technology in Computer Engineering — **CGPA:** 8.6 / 10.0
Publications: AAAI 2020 [5], IEEE CDS [7]

Research Experience

Sony R&D Labs — Zurich, Switzerland
Computer Vision Research Intern — Sep 2023 - Apr 2024
Mentor: Valentina Cavinato

Amazon Science — Washington, USA
Applied Scientist Intern — May 2023 - Aug 2023
Mentor: Dr. Tarang Chugh

Microsoft Research Lab India — Bangalore, India
Research Intern — May 2022 - Aug 2022
Mentors: Dr. Venkat Padmanabhan, Dr. Saumay Pushp

IBM Research Lab — Delhi, India
Research Intern — June 2020 - Aug 2020
Mentor: Dr. Sameep Mehta , Nistha Madaan

Publications

In Progress

[1] **Thinking Deeper Without Growing Larger: Scaling Virtual Depth in Pretrained LLMs for Enhanced Reasoning**
Shivang Chopra, Zhang Xi-Jia, Karmesh Yadav, Yusuf Ali, Chengyue Huang, Zsolt Kira.

Conferences and Journals

[2] **The Geometry of Robustness: Optimizing Loss Landscape Curvature and Feature Manifold Alignment for Robust Finetuning of Vision-Language Models**
Shivang Chopra, Shaunak Halbe, Chengyue Huang, Brisa Maneechotesuwan, Zsolt Kira.
Paper accepted at IEEE/CVF Conference on computer vision and pattern recognition, CVPR 2026

[3] **Refining Text-to-Image Generation: Towards Accurate Training-Free Glyph-Enhanced Image Generation**
Sanyam Lakhanpal, **Shivang Chopra**, Vinija Jain, Aman Chadha, Man Luo
Published at IEEE/CVF Winter Conference on Applications of Computer Vision, WACV 2025

[4] **FRAMES-VQA: Benchmarking Fine-Tuning Robustness across Multi-Modal Shifts in Visual Question Answering**
Chengyue Huang*, Brisa Maneechotesuwan*, **Shivang Chopra**, Zsolt Kira
Published at IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2025

[5] **Hindi-English Hate Speech Detection: Author Profiling, Debiasing, and Practical Perspectives**
Shivang Chopra, Ramit Sawhney, Puneet Mathur, Rajiv Ratn Shah
Published at the AAAI Conference on Artificial Intelligence, AAAI 2020

[6] **Learning to Discern: Imitating Heterogeneous Human Demonstrations with Preference and Representation Learning**
Sachit Kuhar, Shuo Cheng, **Shivang Chopra**, Matthew Bronars, Danfei Xu
Published at the Conference on Robot Learning, CoRL 2023

[7] **Attention-Net: An Ensemble Sketch Recognition Approach using Vector Image**
Shivang Chopra*, Gaurav Jain*, Suransh Chopra*, Anil Singh Parihar
Published in IEEE Transactions on Cognitive and Developmental Systems, IEEE CDS 2022

Workshops/Preprint

[8] **MedMoE: Modality-Specialized Mixture of Experts for Medical Vision-Language Understanding**
Shivang Chopra, Lingchao Mao, Gabriela Sanchez-Rodriguez, Andrew J. Feola, Jing Li, Zsolt Kira
Published at the MMFF-BIOMED workshop at CVPR 2025.

[9] **Source-free domain adaptation with diffusion-guided source data generation**
Shivang Chopra, Suraj Kothawade, Houda Aynaou, Aman Chadha. arXiv 2024

[10] **Active Data Discovery: Mining Unknown Data using Submodular Information Measures**
Suraj Kothawade, **Shivang Chopra**, Saikat Ghosh, Rishabh Iyer
Published at the RealML workshop at ICML 2022.

Patent

[P1] Hybrid Environment for Interactions Between Virtual and Physical Users (**US 18/358,485**), 2024

Awards and Achievements

Graduate Student Scholarship , Georgia Institute of Technology	2023,2024
Research Excellence Award, Delhi Technological University	2020
Student Research Travel Grant, Microsoft Research India	2020
Qualified for Onsite Regionals, ACM ICPC (International Collegiate Programming Contest)	2018

Relevant Coursework

Ph.D. / Master Of Science: Machine Learning, Vision Language Models, Deep Learning for Robotics, Game AI, Natural Language Processing, Learning and Memory, Cognitive Psychology

Bachelor of Technology: Machine Learning, Artificial Intelligence, Natural Language Processing, Distributed Systems, Operating Systems, Computer Networks, Information and Network Security, Computer Organization and Architecture

Technical Skills

Programming Languages	Python, Bash, C, C++
Deep Learning Frameworks	PyTorch, TensorFlow, JAX, Accelerate, Pytorch Lightning
Distributed System Tools	Slurm, Docker, Kubernetes

Community and Academic Services

Reviewer: CVPR 2026, NeurIps 2024-25, ICLR 2025, ICML 2025
Student Volunteer: AAAI 2020, ICRA 2025
Project Lead, AI@GT (Georgia Tech): Mentoring a group of three undergraduate students to build self-evolving embodied agents that learn by interacting with their environment.
Admin, GT Skynet: Part of the team maintaining Georgia Tech's SLURM-based GPU cluster to support ML research and coursework.